

Minhal Rahman

B.Tech Graduate – Applied Electronics & Instrumentation Engineering

☎ +91 6282425867 | ✉ minhalrahman21@gmail.com

🌐 linkedin.com/in/minhalrahman | 🌐 minhalrahman.vercel.app

Summary

Instrumentation and Automation Engineering graduate with hands-on experience in industrial automation, PLC programming, SCADA, HMI, and process control through academic projects and industrial training. Currently pursuing a Professional Diploma in Industrial Automation at IPCS Calicut, with hands-on learning in PLC-based automation and industrial control systems. Experienced in developing automation projects involving CODESYS, Ladder Logic, Structured Text, OPC UA, sensors, actuators, and process monitoring. Interested in automation, control systems, instrumentation, and project engineering, with strong problem-solving skills and a willingness to learn and contribute in dynamic engineering environments.

Education

Government Engineering College

B.Tech, Applied Electronics & Instrumentation Engineering; CGPA: 7.24/10

Kozhikode, Kerala, India

2022 – 2026

IPCS Global

Professional Diploma in Industrial Automation (PDIA)

Calicut, Kerala, India

2026 – Present

- Coursework: PLC, SCADA, HMI, VFD, DCS, Industrial Communication.

Work Experience

SMEC Labs

Industrial Automation Intern

Kozhikode, Kerala, India

May – June 2025

- Gained practical exposure to PLC programming, SCADA systems, and industrial instrumentation.
- Worked with sensors, actuators, and control panels used in industrial automation systems.
- Developed familiarity with PLC-based automation workflows, signal interfacing, and industrial control system fundamentals through hands-on training.

Projects

Smart Warehouse Automation | CODESYS, Ladder Logic, Structured Text, OPC UA, Godot/OIP 2026

- Developed a simulated conveyor-sorting system in which CODESYS controls product identification, classification, tracking, routing, and diverter actuation.
- Implemented a custom CODESYS Product Manager DUT to track each active product's ID, barcode, type, destination, scan encoder value, current position, and sorting status; encoder-based position tracking determines routing based on actual conveyor movement.
- Connected CODESYS with a Godot/OIP 3D digital twin through OPC UA, enabling real-time exchange of sensor, conveyor, and diverter states, with bin-capacity monitoring and warehouse-full interlocks validated in simulation.

Smart Liquid Mixer (ESP32 Based) | ESP32, HTML, CSS, JavaScript

2025

- Developed an ESP32-based automated liquid mixing prototype for process control applications using sensors and actuators.
- Implemented recipe-based sequential mixing logic and process monitoring for automated operation.
- Built a web-based monitoring dashboard using HTML, CSS, and JavaScript for real-time visualization of system parameters.

LabVIEW Simulation of Digital Twin | LabVIEW

2025

- Designed an HMI front panel for real-time monitoring of process level and temperature parameters.
- Developed core process simulation logic with data visualization for monitoring system behavior and operating conditions.

Technical Skills

PLC & Automation: CODESYS, Allen-Bradley RSLogix 500, Ladder Logic, Structured Text, Sequential Control, Interlocks, Timers, Counters, I/O Logic
SCADA & HMI: InTouch, Ignition
Instrumentation & Control: Industrial Instrumentation, Process Control, Temperature Measurement, Level Measurement, Sensors, Actuators
Industrial Communication: OPC UA, TCP/IP
Embedded & Programming: ESP32, Microcontroller Programming, C, MATLAB, HTML, CSS, JavaScript
Engineering Tools: Zelio Soft, LabVIEW, Godot OIP, VS Code, Git/GitHub

Languages

Malayalam: Native Proficiency **English:** Professional Working Proficiency **Hindi:** Elementary Proficiency

Certifications & Workshops

Control Engineering NPTEL, IIT Madras	2025
Siemens Mobility – Commercial Project Manager Job Simulation Forage	2026
Embedded Systems & Automotive Workshop (ENDORA’23) GEC Kozhikode	2023